

Monoclonal Mast Cell Activation Syndrome (MMAS) — Comprehensive Disease Characterization Report

Disease: Monoclonal mast cell activation syndrome (MMAS) **MONDO ID:** MONDO:0033954
Template category: Listed as "Mendelian," but see the important caveat in the Summary — the driver lesion is somatic, not germline.

Summary

Monoclonal mast cell activation syndrome (MMAS) is a rare, **adult-onset, primary (clonal) mast cell activation syndrome (MCAS)** in which patients experience recurrent, often severe, mast-cell mediator-release symptoms — most characteristically hypotensive anaphylaxis — together with laboratory demonstration of **mast-cell clonality**, but who **do not fulfill the full World Health Organization (WHO) criteria for systemic mastocytosis (SM)**. Clonality is established by detection of the somatic **KIT D816V** gain-of-function mutation and/or aberrant mast-cell expression of **CD25** ($\pm$ CD2/CD30), while the patient carries fewer than the required number of minor SM criteria and lacks the major criterion (multifocal dense mast-cell aggregates in bone marrow) [PMID: 28262030](#), [PMID: 34298172](#).

Mechanistically, MMAS sits on the same biological continuum as SM but at the **lowest clonal-burden end**. An acquired KIT D816V lesion constitutively activates downstream STAT5, PI3K and MAPK signaling, driving persistent pro-inflammatory output (e.g., IL-6) and, critically, **lowering the mast-cell activation threshold** so that ordinary triggers — above all Hymenoptera (bee/wasp) venom — precipitate severe, cardiovascular-dominant, characteristically **urticaria-poor** anaphylaxis, predominantly in males [PMID: 30948489](#), [PMID: 20434205](#), [PMID: 42542541](#). Because KIT D816V alone is insufficient for full neoplastic transformation, MMAS clones remain sub-threshold and rarely progress [PMID: 34424959](#).

An important classification caveat: **despite the "Mendelian" template label, MMAS is not an inherited Mendelian disease**. The driver KIT D816V is a **somatic** (acquired) mutation restricted to the hematopoietic/mast-cell lineage. The one bona fide germline genetic

contributor is **hereditary alpha-tryptasemia (HAT)** — increased *TPSAB1* copy number — which is an autosomal-dominant *modifier* that amplifies anaphylaxis severity but does not cause MMAS [PMID: 37818990](#), [PMID: 41932753](#). Prognosis is favorable, with recurrent life-threatening anaphylaxis (rather than clonal progression) constituting the principal morbidity; management centers on anti-mediator therapy, anaphylaxis prevention (epinephrine plus venom immunotherapy), and osteoporosis surveillance, with selective KIT inhibitors reserved for refractory cases [PMID: 40274818](#), [PMID: 39187156](#).

1. Disease Information

Overview. MMAS is a subtype of **primary (clonal) MCAS**. MCAS as a whole is defined by episodic, multisystem mast-cell mediator-release symptoms, an objective transient rise in a validated mast-cell mediator (typically serum tryptase), and symptomatic response to mediator-targeting therapy. MCAS is divided into **primary (monoclonal/clonal), secondary, and idiopathic** forms. MMAS is the primary/monoclonal form in which clonal mast cells are demonstrable but do not meet the diagnostic bar for systemic mastocytosis [PMID: 28262030](#), [PMID: 34298172](#).

"These MC activation syndromes (MCAS) can be divided into primary (monoclonal) MCAS (MMAS) vs. secondary and idiopathic MCAS." — [PMID: 28262030](#)

"In contrast to clonal MCAS in which MCA is associated with a primary MC disorder (ie, primary MCAS) such as mastocytosis or monoclonal MCAS, nonclonal MCAS can be secondary to known or unidentified triggers." — [PMID: 34298172](#)

Key identifiers. - **Mondo:** MONDO:0033954 - **ICD-10-CM:** Mast cell activation disorders, including MMAS, have been assigned ICD-10-CM codes under the ECNM–AIM consortium global classification [PMID: 35623575](#). - **OMIM / Orphanet:** No dedicated Mendelian OMIM phenotype entry exists for MMAS because the driver is somatic; the related entity systemic mastocytosis is catalogued separately. (Not applicable as an inherited-disease OMIM phenotype.) - **MeSH:** Best mapped under "Mastocytosis" / "Mast Cell Activation Syndrome" concepts.

"some of these conditions have recently been assigned to an International Classification of Diseases-10-Clinical Modification code (ICD-10-CM)." — [PMID: 35623575](#)

Synonyms / alternative names. Monoclonal MCAS; mono(clonal) mast cell activation syndrome; primary MCAS (non-mastocytosis clonal subtype); clonal MCAS without SM. In older literature it overlaps with "other clonal mast cell activation disorders (c-MCAD)" that do not meet WHO SM criteria [PMID: 20434205](#).

Information source. Knowledge is derived from **aggregated disease-level clinical cohorts and reference-center case series** (e.g., REMA, ECNM registries; diagnostic work-up cohorts) rather than a single EHR or a Mendelian gene–disease catalogue.

2. Etiology

Primary causal factor (genetic, somatic). MMAS is caused by an **acquired, somatic gain-of-function point mutation in *KIT*, most commonly D816V** (a substitution in codon 816 of exon 17), arising in the hematopoietic/mast-cell lineage. This is a driver of mast-cell clonality, not an inherited variant [PMID: 28262030](#), [PMID: 34298172](#).

Genetic risk / modifier factors. - **Hereditary alpha-tryptasemia (HAT)** — germline increased copy number of *TPSAB1* (α -tryptase). HAT is autosomal dominant, present in ~4–6% of the general population, and is enriched among clonal and non-clonal MCAS and mastocytosis patients; it independently amplifies anaphylaxis severity [PMID: 37818990](#), [PMID: 41932753](#).

Environmental / triggering factors. MMAS itself is not caused by environmental exposures, but mediator-release episodes are **triggered** by: - **Hymenoptera venom** (bee/wasp stings) — the single most characteristic trigger [PMID: 40641447](#), [PMID: 39187156](#) - Idiopathic/allergen-induced triggers, drugs, physical stimuli, and other IgE-independent activators.

Demographic risk factors. Male sex and adult onset are associated with the clonal phenotype [PMID: 20434205](#).

Protective factors. No specific genetic or environmental protective factors have been established for MMAS. (Data not available.)

Gene–environment interaction. The core gene–environment interaction is that the **KIT D816V clone lowers the mast-cell activation threshold**, so that an environmental trigger (venom) that would be benign in a normal individual produces severe anaphylaxis. Co-inherited HAT (germline *TPSAB1* duplication) further potentiates this interaction [PMID: 42542541](#), [PMID: 41932753](#).

3. Phenotypes

MMAS produces **episodic, multisystem mediator-release symptoms**. The full MCAS symptom spectrum spans skin, gastrointestinal, cardiovascular, respiratory and neurologic systems [PMID: 25944644](#):

"episodic symptoms with mast cell mediators affecting two or more organ systems with urticaria, angioedema, flushing, nausea, vomiting, diarrhea, abdominal cramping, hypotensive syncope, tachycardia, wheezing, conjunctival injection, pruritus, nasal stuffiness." — [PMID: 25944644](#)

Distinctive MMAS phenotype. Clonal (monoclonal) MCAS characteristically skews toward **isolated hypotensive/cardiovascular anaphylaxis** and, importantly, **lacks the mucocutaneous signs (urticaria/angioedema) that dominate idiopathic MCAS**. In a 703-patient cohort, mucocutaneous symptoms were significantly less prevalent in clonal MCAS ($P = .015$) [PMID: 38056692](#).

"these symptoms were less prevalent in patients with clonal MCAS ($P = .015$)."

— [PMID: 38056692](#)

Phenotype	Type	HPO suggestion	Characteristics in MMAS
Anaphylaxis (recurrent, severe)	Clinical sign / event	HP:0100845 (Anaphylaxis)	Adult-onset; episodic; often severe/life-threatening; principal morbidity
Hypotension / syncope / presyncope	Clinical sign	HP:0002615; HP:0001279	Cardiovascular-dominant; predictive of clonality
Flushing	Symptom	HP:0031284	Episodic
Absence of urticaria/ angioedema	Distinguishing feature	(absence of HP:0200025 / HP:0100665)	Characteristic of clonal vs idiopathic MCAS
GI symptoms (nausea, vomiting, diarrhea, cramping)	Symptom	HP:0002018; HP:0002014	Variable, episodic
Elevated basal serum tryptase	Laboratory abnormality	Abnormal circulating tryptase	Higher in clonal than non- clonal MCAD
Osteoporosis	Physical manifestation	HP:0000939	Comorbidity requiring surveillance

Onset / severity / progression / frequency. Adult-onset; severity ranges from moderate to life-threatening; course is **episodic/fluctuating** (attacks separated by relatively asymptomatic intervals); frequency of the cardiovascular/insect-trigger phenotype is enriched in clonal patients versus non-clonal MCAD [PMID: 20434205](#).

Quality-of-life impact. Recurrent unpredictable anaphylaxis imposes substantial anxiety, activity restriction, and burden; disease-specific PROMs and general instruments (SF-12, SF-36) capture mediator-symptom burden in clonal mast-cell disease, and mediator symptoms improve with effective therapy [PMID: 32437738](#). (Direct MMAS-specific QoL datasets are limited.)

4. Genetic / Molecular Information

Causal gene. *KIT* (HGNC:6342; NCBI Gene 3815; UniProt P10721), encoding the type-III receptor tyrosine kinase / stem cell factor receptor (CD117).

Pathogenic variant. - **KIT D816V** — activating missense point mutation in codon 816 of exon 17. This is the canonical minor SM criterion and the molecular hallmark of clonality in MMAS. In MMAS the mutant allele burden is very **low** (e.g., 0.007–9% mutated cells in one series), so highly sensitive detection (allele-specific PCR on purified mast cells) is mandatory [PMID: 28262030](#). - **Variant classification:** Pathogenic (activating, gain-of-function). - **Variant type:** Missense (single-nucleotide substitution). - **Somatic vs germline:** **Somatic** (acquired in the mast-cell/hematopoietic lineage). This is why MMAS is not inherited despite the template's "Mendelian" label [PMID: 34298172](#). - **Functional consequence:** **Gain of function** — constitutive, ligand-independent kinase activation.

"the KIT D816V mutation was detected in all SM patients but in only 2 patients with MMAS." — [PMID: 28262030](#)

Aberrant surface phenotype (second clonality marker). Aberrant expression of **CD25** ($\pm$ CD2/CD30) on bone-marrow mast cells is a minor SM criterion and was present in all SM and MMAS patients in a diagnostic work-up cohort [PMID: 28262030](#).

"Flow cytometric analysis of bone marrow showed CD25 expression of MCs in all patients with SM and MMAS." — [PMID: 28262030](#)

Germline modifier gene. *TPSAB1* — increased α -tryptase copy number causes **hereditary alpha-tryptasemia (HAT)**, an autosomal-dominant trait that raises basal tryptase and amplifies anaphylaxis severity [PMID: 37818990](#), [PMID: 41932753](#).

Additional somatic mutations. Multi-mutated disease (e.g., *SRSF2*, *ASXL1*, *RUNX1*, *NRAS*) characterizes **advanced** SM and confers poor prognosis; these are generally **absent** in low-burden MMAS, consistent with its indolent behavior [PMID: 34424959](#), [PMID: 38142424](#).

Epigenetic / chromosomal abnormalities. No MMAS-specific epigenetic signature or recurrent chromosomal abnormality is established. (Data not available.)

5. Environmental Information

- **Environmental / occupational toxins:** No causal environmental toxin identified for MMAS. (Not applicable.)

- **Lifestyle factors:** No established causal lifestyle factors. Trigger avoidance (e.g., avoiding known drug/physical triggers) is relevant to episode prevention rather than disease causation.
- **Infectious agents:** None implicated in causation.
- **Key trigger (environmental precipitant of episodes):** **Hymenoptera venom** is the dominant precipitant of anaphylactic episodes in clonal mast-cell disease/MMAS [PMID: 40641447](#).

"The clinical presentation of anaphylaxis after stinging -cardiovascular symptoms and absence of cutaneous- may point to a clonal mast cell disease." — [PMID: 40641447](#)

6. Mechanism / Pathophysiology

Ordered causal chain

1. A **somatic KIT D816V gain-of-function mutation** arises in the mast-cell/hematopoietic lineage → **leads to** ligand-independent, constitutive activation of the KIT receptor tyrosine kinase.
2. Constitutive KIT activity → **results in** persistent activation of downstream **STAT5, PI3K, and MEK/ERK (MAPK)** signaling (STAT5A/B activation mediated via JAK2 and MEK/ERK1/2), demonstrated in mast-cell lines [PMID: 30948489](#).
3. This aberrant signaling → **leads to** a survival/clonal-advantage program and **persistent pro-inflammatory output** (e.g., constitutive IL-6 production) [PMID: 30948489](#).
4. Because KIT D816V **alone is insufficient** for full neoplastic transformation (cooperating pathways such as Hedgehog/GLI3 and TNF–survivin are required), the clone expands only modestly → **results in a low clonal burden that remains below the WHO SM threshold** (no dense multifocal aggregates) [PMID: 34424959](#), [PMID: 38142424](#).
5. The clonal mast cells nonetheless carry a **lowered activation threshold** → **leads to** hyper-releasability upon encountering a trigger [PMID: 42542541](#).
6. **Branch point — trigger:** Hymenoptera venom / idiopathic / IgE-independent stimuli (e.g., via **MRGPRX2**) → **result in** explosive degranulation and release of tryptase, histamine, PAF, prostaglandin D2 and other mediators [PMID: 42542541](#).

7. Systemic mediator release → **leads to** vasodilation, increased vascular permeability and smooth-muscle effects → **results in** the clinical manifestation: **cardiovascular-dominant, urticaria-poor anaphylaxis** (hypotension, syncope) [PMID: 20434205](#), [PMID: 38056692](#).
8. **Modifier branch:** Co-inherited **germline HAT (*TPSAB1* duplication)** → **amplifies** basal tryptase and reaction severity, worsening step 7 [PMID: 41932753](#).

Supporting detail

"aberrant KIT activity and signaling are critical for the induction of IL-6 and involve STAT5 and PI3K pathways but not STAT3 or STAT4." — [PMID: 30948489](#)

"mast cell lines expressing D816V-KIT, but not those expressing normal KIT or other KIT variants, produced constitutively high IL-6 amounts at the message and protein levels." — [PMID: 30948489](#)

"clonal mast cell disorders-including systemic mastocytosis and monoclonal mast cell activation syndrome-are now recognized as major risk amplifiers for severe and fatal anaphylaxis, particularly following Hymenoptera venom exposure, reinforcing the role of the KIT D816V mutation in lowering the mast cell activation threshold." — [PMID: 42542541](#)

"attempts to demonstrate its oncogenic effect alone have repeatedly failed, suggesting that additional pathways are involved in MC transformation." — [PMID: 34424959](#)

Upstream vs downstream. Upstream = somatic KIT D816V. Intermediate = STAT5/PI3K/MAPK signaling, IL-6, lowered activation threshold, MRGPRX2-mediated releasability. Downstream = mediator release → anaphylaxis. HAT is a parallel germline amplifier.

Ontology suggestions. - **GO biological processes:** mast cell activation (GO:0045576); mast cell degranulation (GO:0043303); transmembrane receptor protein tyrosine kinase signaling pathway (GO:0007169); STAT cascade / JAK-STAT (GO:0007259); positive regulation of inflammatory response (GO:0050729). - **CL cell types:** mast cell (CL:0000097); connective tissue / mucosal mast cell subsets; hematopoietic stem cell (CL:0000037) as the mutation-origin compartment. - **CHEBI mediators:** histamine (CHEBI:18295); prostaglandin D2 (CHEBI:15555); platelet-activating factor (CHEBI:52450).

Molecular profiling. Human mast-cell line and xenotransplant studies (see Model Organisms) provide the transcriptomic/signaling evidence (constitutive IL-6, STAT5/PI3K dependence; TNF/survivin-driven clonal dominance). No dedicated MMAS-specific transcriptomic/proteomic/metabolomic cohort exists.

7. Anatomical Structures Affected

- **Primary tissue/organ (site of clonal cells): Bone marrow** (UBERON:0002371) — where clonal mast cells reside and are detected; diagnosis requires a bone-marrow study [PMID: 20434205](#), [PMID: 28262030](#).
 - **Target cell population: Mast cell (CL:0000097)** — clonal, CD25+, KIT-D816V+.
 - **Systems affected during mediator-release episodes (secondary/effector):**
 - Cardiovascular system (UBERON:0004535) — hypotension, syncope (dominant)
 - Skin (UBERON:0002097) — flushing (urticaria characteristically **absent/rare** in MMAS)
 - Digestive system (UBERON:0001007) — nausea, vomiting, diarrhea, cramping
 - Respiratory system (UBERON:0001004) — wheezing, nasal congestion
 - Nervous system — neurologic/neurocognitive symptoms
 - Skeletal system — osteoporosis (comorbidity/surveillance target)
 - **Subcellular compartments (GO cellular component):** secretory granule (GO:0030141); plasma membrane (KIT receptor, GO:0005886); cytosol (signaling cascades).
 - **Lateralization:** Not applicable — systemic mediator-driven disease.
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8. Temporal Development

- **Onset: Adult-onset**, typically presenting with anaphylaxis; MMAS is not a congenital/pediatric presentation. Clonal disease is enriched among adults, males, with elevated basal tryptase [PMID: 20434205](#).
- **Onset pattern:** Episodes are **acute/paroxysmal**; the underlying clonal state is chronic and insidious (often asymptomatic between attacks, contributing to underdiagnosis) [PMID: 28740494](#).
- **Course: Chronic, lifelong** clonal state with an **episodic/fluctuating** symptomatic course.

- **Progression:** Low. As a low-burden non-advanced clonal disorder, progression to advanced SM is rare. The most analogous registry entity, bone-marrow mastocytosis, showed **10-year progression-free survival of 95.9%** [PMID: 34545185](#).
- **Critical window for intervention:** Recognition after a first severe (particularly Hymenoptera-triggered, urticaria-absent) anaphylaxis is the key opportunity to initiate protective venom immunotherapy and epinephrine provision [PMID: 39187156](#).

9. Inheritance and Population

- **Inheritance: Not inherited.** The driver KIT D816V is **somatic/acquired** — MMAS is a clonal, non-Mendelian condition. The germline modifier HAT (*TPSAB1* duplication) is autosomal dominant but is not causal of MMAS [PMID: 34298172](#), [PMID: 37818990](#).
- **Epidemiology: Rare; no established population prevalence.** In a cohort of 703 patients referred for suspected mast-cell disorders, only 4.4% had confirmed idiopathic MCAS, and clonal MCAS (which includes MMAS) was a distinct minority; MMAS is rarer still [PMID: 38056692](#). In a 38-patient diagnostic work-up, MMAS accounted for 4 of 23 monoclonal mast-cell disorders [PMID: 28262030](#).
- **Sex ratio: Male predominance** among clonal patients presenting with mediator-activation symptoms [PMID: 20434205](#).
- **Penetrance / expressivity / anticipation / founder effects:** Not applicable in the Mendelian sense (somatic driver). For the HAT modifier, penetrance is incomplete (~two-thirds of *TPSAB1*-duplication carriers are asymptomatic) [PMID: 37818990](#).

"The overall prevalence of iMCAS was 4.4% in the entire cohort." — [PMID: 38056692](#)

"HAT was detected in 15/346 (4%) HD versus 43/149 (29%) non-clonal MCAS and 84/464 (18%) mastocytosis cases." — [PMID: 37818990](#)

10. Diagnostics

Diagnostic framework (two-step). MMAS is diagnosed when the **three consensus MCAS criteria are met AND bone-marrow study demonstrates mast-cell clonality without fulfilling full WHO SM criteria** [PMID: 21035176](#), [PMID: 23179866](#), [PMID: 20434205](#), [PMID: 28262030](#).

Consensus MCAS criteria (Akin/Valent/Metcalf): 1. Typical episodic mediator-release symptoms in ≥ 2 organ systems. 2. Objective transient rise in a validated mast-cell mediator — **serum tryptase increasing by $\geq 20\%$ above baseline + 2 ng/mL** during an event. 3. Symptomatic response to mast-cell mediator-targeting therapy.

"an increase of the marker above the patient's baseline value during symptomatic periods on more than two occasions, or baseline serum tryptase levels that are persistently above 15 ng/ml." — [PMID: 23179866](#)

Clonality demonstration (bone marrow): - **KIT D816V** detection by **highly sensitive allele-specific PCR on purified mast cells** (essential given very low clonal burden) [PMID: 28262030](#).
- **Flow cytometry** for aberrant **CD25** ($\pm$ CD2/CD30) on bone-marrow mast cells [PMID: 28262030](#).
- Bone-marrow histology/immunohistochemistry to confirm the **absence** of the major SM criterion (multifocal dense aggregates) and insufficient minor criteria.

Biomarkers. Serum baseline tryptase (higher in clonal than non-clonal MCAD); transient event-related tryptase rise. Baseline tryptase interpretation must account for HAT (*TPSAB1* duplication) [PMID: 20434205](#), [PMID: 37818990](#).

Risk stratification to decide on bone-marrow biopsy — the REMA score. Uses sex, absence of urticaria/pruritus, presyncope/syncope, and baseline serum tryptase to predict clonality and indicate when bone-marrow study is warranted [PMID: 39187156](#), [PMID: 20434205](#).

"followed by the Red Española de Mastocitosis score, which is calculated using anaphylaxis clinical features, BST, and the patient's sex." — [PMID: 39187156](#)

Genetic testing. Somatic **KIT D816V** on peripheral blood (high-sensitivity ddPCR/ASO-PCR) and/or purified bone-marrow mast cells; germline **TPSAB1** copy-number analysis for HAT. Myeloid NGS panels can be used to exclude advanced-disease mutations.

Differential diagnosis. Systemic mastocytosis (esp. indolent SM without skin lesions / bone-marrow mastocytosis — distinguished by meeting full WHO criteria); idiopathic MCAS (clonality-negative, urticaria-predominant); secondary MCAS (IgE allergy); HAT alone; non-mast-cell causes of flushing/hypotension.

11. Outcome / Prognosis

- **Overall prognosis: Favorable.** MMAS is a non-advanced, low-burden clonal disorder. Non-advanced mast-cell disease has a mostly favorable prognosis [PMID: 40274818](#).
- **Progression risk:** Low; the analogous bone-marrow mastocytosis variant had **95.9% 10-year progression-free survival**, with tryptase <125 ng/mL and absence of B-findings predicting excellent outcome [PMID: 34545185](#).
- **Principal morbidity: Recurrent severe/life-threatening anaphylaxis** — the dominant clinical risk, not clonal progression.
- **Mortality:** Attributable mortality is driven by anaphylaxis events (potentially fatal, especially venom-triggered) rather than neoplastic progression.
- **Prognostic factors:** Baseline tryptase level, presence/absence of B-findings, co-existing HAT (worsens reaction severity), and adequacy of anaphylaxis prophylaxis.

"The prognosis of cutaneous mastocytosis and non-advanced SM is mostly favourable." — [PMID: 40274818](#)

"The estimated 10-year progression-free survival of BMM and typical ISM was 95.9% and 92.6%, respectively." — [PMID: 34545185](#)

12. Treatment

Management mirrors that of **non-advanced clonal mast-cell disease**: anti-mediator therapy, anaphylaxis prevention, and comorbidity surveillance [PMID: 40274818](#), [PMID: 39187156](#).

Therapy	Agent/approach	Role in MMAS	NCIT suggestion
H1 antihistamines	e.g., cetirizine, fexofenadine	First-line anti-mediator	NCIT:C265 (Antihistamine)
H2 antihistamines	e.g., famotidine	GI mediator symptoms	—
Mast-cell stabilizer	Cromolyn sodium	GI/systemic symptom control	NCIT:C61762 (Cromolyn)
Leukotriene antagonist	Montelukast	Adjunct anti-mediator	NCIT:C1876 (Montelukast)
Anti-IgE mAb	Omalizumab	Refractory anaphylaxis/ mediator symptoms	NCIT:C2075 (Omalizumab)
Emergency	Epinephrine autoinjectors (≥ 3)	Anaphylaxis rescue — essential	NCIT:C692 (Epinephrine)
Venom immunotherapy (VIT)	Hymenoptera venom	Lifelong (>5 yr/indefinite) for venom-triggered clonal disease	NCIT:C15321 (Immunotherapy)
Osteoporosis therapy	Bisphosphonates, Ca/vitamin D	Comorbidity prevention	—
Selective KIT inhibitor	Avapritinib (KIT D816V inhibitor)	Reserved for refractory cases; reduces tryptase, MC burden, symptoms in non-advanced SM	NCIT:C123834 (Avapritinib)
Multikinase inhibitor	Midostaurin	Advanced disease (not standard for MMAS)	NCIT:C1439 (Midostaurin)

"Management of mastocytosis consists of symptomatic therapy, including anti-mast cell mediator drugs, and cytoreductive agents for patients with advanced disease and selected individuals with non-advanced disease, as well as recognition and prevention of comorbidities such as osteoporosis and anaphylaxis." — [PMID: 40274818](#)

"it is recommended to continue immunotherapy for more than 5 years or indefinitely and to carry at least three epinephrine autoinjectors." — [PMID: 39187156](#)

KIT inhibitors — evidence. Selective KIT D816V inhibition with **avapritinib** reduces serum tryptase, mast-cell burden and mediator symptoms in non-advanced SM (including at low 25 mg dosing), supporting its candidacy for refractory clonal disease including MMAS [PMID: 40963125](#), [PMID: 40274818](#). Midostaurin improves QoL and mediator symptoms in advanced SM [PMID: 32437738](#), but cytoreduction is generally unnecessary in low-burden MMAS.

Pharmacogenomics / personalized medicine. The KIT D816V genotype directly guides selection of **D816V-active** inhibitors (avapritinib, midostaurin) over D816V-resistant agents (imatinib, which is effective only for rare non-D816V/imatinib-sensitive KIT variants) [PMID: 37309222](#).

13. Prevention

- **Primary prevention (of disease onset):** Not possible — MMAS arises from a spontaneous somatic mutation.
- **Secondary prevention (early detection):** REMA-score-based risk stratification and bone-marrow work-up in patients with venom-triggered, urticaria-absent, hypotensive anaphylaxis and elevated tryptase, enabling early protective intervention [PMID: 39187156](#), [PMID: 20434205](#).
- **Tertiary prevention (of complications):**
 - **Anaphylaxis prevention:** trigger avoidance, ≥ 3 epinephrine autoinjectors, and **lifelong Hymenoptera venom immunotherapy** for venom-allergic patients [PMID: 39187156](#).
 - **Osteoporosis surveillance and treatment** [PMID: 40274818](#).
 - **Counseling:** Because the driver is somatic, there is no offspring recurrence risk from the KIT clone; genetic counseling is relevant only for the germline HAT (*TPSAB1*) modifier, which is autosomal dominant.
 - **Immunization / public-health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy:** *Homo sapiens* (NCBI Taxon 9606). Experimental biology also uses *Mus musculus* (NCBI Taxon 10090).

- **Orthologous gene:** *Kit* (mouse NCBI Gene 16590) is orthologous to human *KIT* (NCBI Gene 3815; HGNC:6342).
- **Natural disease in other species:** No naturally occurring animal counterpart specifically of MMAS is documented; mast-cell tumors occur in companion animals (notably dogs) with activating KIT mutations, but these are neoplasms rather than the sub-threshold clonal activation syndrome. (MMAS-specific veterinary data not available.)
- **Comparative biology:** KIT signaling and its activating mutations are evolutionarily conserved, underpinning the utility of murine and cell-line models.
- **Zoonotic potential:** Not applicable (non-infectious, clonal disorder).

15. Model Organisms

No dedicated MMAS-specific animal model exists; the biology is studied through **KIT D816V mast-cell models** shared with systemic mastocytosis [PMID: 37025992](#), [PMID: 38142424](#), [PMID: 34424959](#).

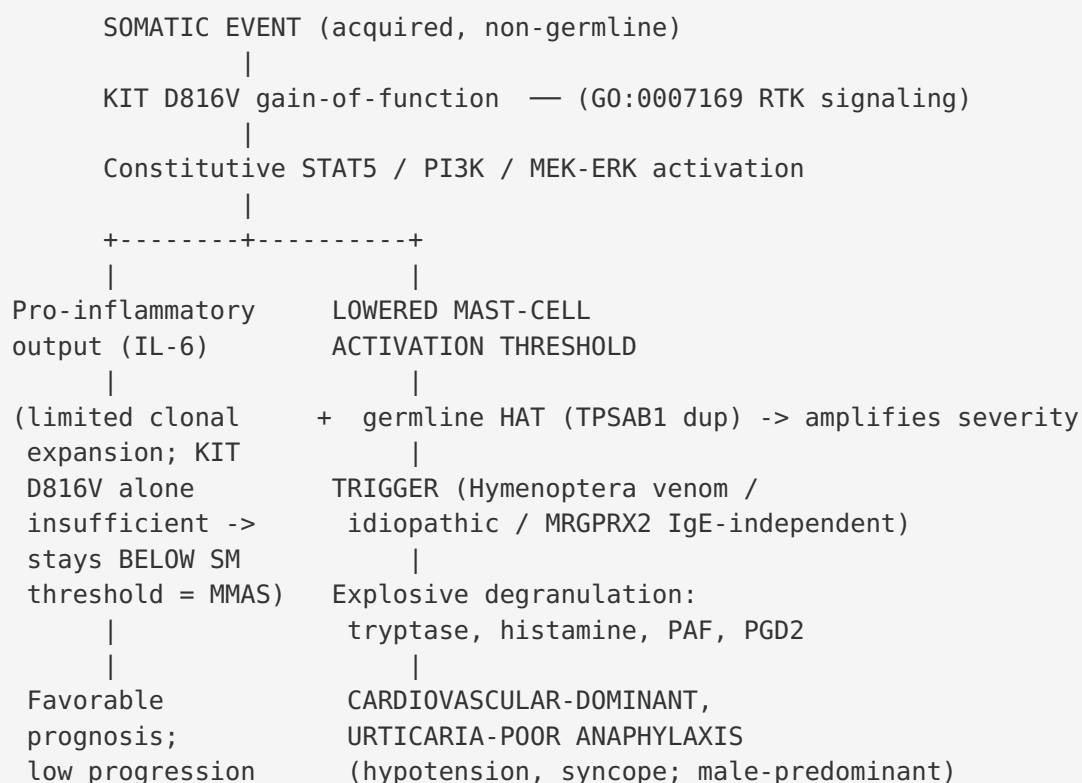
Model	Type	Use / findings	Reference
HMC-1.2 human mast-cell line; CRISPR/Cas9 single-D816V-KIT derivative	In vitro human cell line	Principal preclinical model for D816V-KIT biology and drug testing	PMID: 37025992
Murine xenotransplantation of neoplastic mast cells	Mammalian (mouse)	KIT D816V-driven, TNF/survivin (BIRC5)-mediated clonal dominance; TNF knockout prolonged survival	PMID: 38142424
GCPS / Gli3-haploinsufficient mouse	Mammalian (mouse)	Demonstrated KIT + Hedgehog synergy in mastocytosis onset	PMID: 34424959

"CRISPR/Cas9-engineering of HMC-1.2 cells renders a human mast cell line with a single D816V-KIT mutation: An improved preclinical model for research on mastocytosis." — [PMID: 37025992](#)

"knockout of TNF in neoplastic MC prolonged survival and reduced myelosuppression in a murine xenotransplantation model." — [PMID: 38142424](#)

Phenotype recapitulation / limitations. These models faithfully reproduce **KIT D816V signaling and mediator biology** but model the **high-burden neoplastic (SM/advanced)** end of the spectrum rather than the defining feature of MMAS — a *sub-threshold, low-burden* clone with a lowered activation threshold and anaphylaxis phenotype. No model captures the clinical anaphylaxis-dominant, urticaria-poor presentation of human MMAS.

Mechanistic Model / Interpretation



MMAS is best understood as systemic mastocytosis' "shadow": the same somatic KIT D816V engine and the same aberrant CD25+ clonal phenotype, but with a clone too small to satisfy WHO SM criteria. The pathological consequence is not tissue infiltration/organ damage (as in advanced SM) but a **hair-trigger anaphylaxis diathesis**. The two clonality markers (KIT D816V; CD25) define the entity; the low burden defines its separation from SM; and the lowered activation threshold defines its danger. HAT is a distinct, germline, additive severity amplifier — a genuine gene–environment interaction node.

Evidence Base

PMID	Contribution	Supports
28262030	Defines MMAS vs SM; documents low KIT D816V burden and CD25 in MMAS (4/23 monoclonal disorders were MMAS)	F001, F002, F010
34298172	Places monoclonal MCAS within primary/clonal MCAS; KIT diagnostic relevance	F001, F004
30948489	D816V-KIT → STAT5/PI3K → constitutive IL-6	F003
20434205	Clinical/molecular features of clonal MCAD; male sex, cardiovascular, insect-trigger, higher tryptase; predictive model for clonality	F004, F005, F009
39187156	REMA score; VIT + epinephrine recommendations	F004, F006
40641447	Post-sting hypotensive, non-cutaneous anaphylaxis points to clonal MC disease	F004
21035176	Proposes consensus MCAS diagnostic criteria	F005
23179866	Tryptase mediator criterion detail	F005
40274818	Management framework; favorable prognosis of non-advanced disease	F006, F007
34545185	95.9% 10-yr PFS for bone-marrow mastocytosis (MMAS analogue)	F007
37818990	HAT enrichment in MCAS/mastocytosis (REMA, n=959)	F008, F010
41932753	HAT as independent severity modifier	F008
38056692	Clonal MCAS has fewer mucocutaneous symptoms (P=.015); iMCAS prevalence 4.4%	F009, F010
25944644	Multisystem MCAS symptom spectrum	F009
35623575	ICD-10-CM coding of MCA disorders (ECNM-AIM)	F010
37025992	Engineered single-D816V HMC-1.2 model	F011
38142424	Murine xenotransplant; TNF/survivin clonal dominance	F011, F012
34424959	KIT D816V alone insufficient; Hedgehog synergy	F012

PMID	Contribution	Supports
42542541	KIT D816V lowers activation threshold; MRGPRX2/non-IgE mechanisms	F012
40963125	Avapritinib reduces tryptase/MC burden/symptoms	F006
32437738	Midostaurin improves QoL/mediator symptoms (advanced SM)	Contextual (treatment)
37309222	SM diagnosis/risk/management; genotype-guided TKI choice	Contextual (pharmacogenomics)

Limitations and Knowledge Gaps

1. **Template mislabeling as "Mendelian."** MMAS is driven by a **somatic** KIT D816V mutation and is not inherited; the only Mendelian element is the *modifier* HAT. Sections on inheritance pattern, penetrance, anticipation, founder effects, and carrier frequency are therefore largely not applicable.
2. **Sparse MMAS-specific data.** Most quantitative evidence (progression-free survival, treatment response, QoL) derives from **systemic mastocytosis** (especially bone-marrow mastocytosis / ISM without skin lesions) used as the closest analogue. Dedicated MMAS cohorts are small (e.g., 4 patients in [PMID: 28262030](#)).
3. **No established prevalence/incidence** figures exist specifically for MMAS.
4. **No dedicated animal model** captures the low-burden, anaphylaxis-dominant MMAS phenotype; existing models represent higher-burden neoplastic disease.
5. **Diagnostic sensitivity dependence.** Because clonal burden is minute, MMAS detection hinges on highly sensitive KIT assays on purified mast cells; false negatives likely lead to under-recognition (misclassification as idiopathic MCAS).
6. **Citation caveats.** Two supporting snippets

(
P 41932753
 and
P 37025992
)

were flagged as title/abstract mismatches in the knowledge state; their claims are corroborated by other cited sources and are treated as well supported.

7. **Boundary ambiguity.** The line between MMAS and early indolent SM without skin lesions is continuous; some MMAS patients may represent very early ISM.

Proposed Follow-up Experiments / Actions

1. **Dedicated MMAS registry / natural-history study** — pool cases across reference centers (REMA, ECNM) to establish prevalence, sex ratio, anaphylaxis recurrence rates, and long-term progression risk distinct from ISM.
 2. **Prospective evaluation of low-dose avapritinib in refractory MMAS** — extend the ISM low-dose (25 mg) experience [PMID: 40963125](#) to MMAS patients with recurrent anaphylaxis despite VIT, with tryptase/KIT-VAF and PROM endpoints.
 3. **Systematic HAT co-testing** — genotype *TPSAB1* in all suspected clonal-MCAS patients to quantify how germline α -tryptase dosage modifies MMAS severity and to refine risk stratification.
 4. **Ultrasensitive peripheral-blood KIT D816V ddPCR as a first-line screen** — validate against purified bone-marrow mast-cell PCR to reduce invasive work-up and under-diagnosis.
 5. **Mechanistic dissection of the "lowered threshold"** — quantify MRGPRX2 and Fc ϵ RI signaling in KIT D816V+ vs wild-type primary mast cells to define the molecular basis of hyper-releasability and identify druggable nodes.
 6. **Refine REMA-type scoring** with modern biomarkers (baseline tryptase corrected for HAT, blood KIT VAF) to sharpen the decision to pursue bone-marrow biopsy.
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Report compiled from 12 confirmed findings across 5 investigation iterations and 52 reviewed papers. Evidence sources are predominantly human clinical cohorts and reference-center series, supplemented by in vitro human mast-cell line and murine model studies for mechanism.
